Antoine Montagne

Data Scientist

antoinemontagne387@gmail.com

06 46 88 40 90

Île-de-France

After nearly two years in data consulting, where I built strong ML and DevOps skills, I am now seeking a Data Scientist role in a dynamic environment and am available to start immediately.

Education

De septembre 2022 à février 2024

Post-Master's Degree in Big Data

Télécom Paris Palaiseau

- **Statistics**: Ordinary Least Squares, Statistical hypothesis testing, Time series & Autoregressive models
- **Machine Learning**: Random Forest, SVM, Dimension Reduction, Ensemble learning
- **Deep Learning**: CNN, RNN, Hidden Markov model, Graph Neural Network, NLP and LLM (BERT), Generative models
- **Web**: Scraping, Web development and APIs

De septembre 2019 à août 2022

Engineering degree in Microelectronics and Computing

École Nationale Supérieure des Mines de Saint-Etienne Gardanne

- lst year: Networks, Graphs & Optimization, Assembler & C, Electronics
- 2nd year: SQL, Microcontroller system, Cryptography, C++
- Academic semester in Hamburg: Machine Learning, Network Optimization (Markov chains, Queuing theory), Cybersecurity, Network Protocols

Experiences

Depuis octobre 2024

MPData, Data Scientist Consultant, Technical Assistance

Air France Roissy Charles de Gaulle

- Developed ML forecasting models for two projects : **ground staff absenteeism** and **lounge attendance** at CDG. Improving workforce planning, reducing temporary staffing costs and optimizing food supply management.
- Reduced **MAPE** model from **5%** through Optuna hyperparameter tuning and advanced feature engineering.
- Migrated pipelines from on-premise to **GCP** (GCS, BigQuery, Vertex AI Pipelines), enhancing scalability and maintainability.
- Processed tens of millions of records with PySpark on Dataproc Serverless, **reducing preprocessing time** from **1 hour to 30 minutes** while lowering compute costs.
- Automated model deployment using **GitHub Actions** and built a Spotfire **dashboard** to monitor performance/data drift.

De juin 2024 à septembre 2024

MPData, Data Scientist Consultant, Fixed-Price Project

Safran SAE Boulogne-Billancourt

- Built a Python-based Machine Learning model to **simulate aircraft engine torque behavior**, reducing reliance on physical test benches and lowering testing costs.
- Integrated the Python model into the client's **MATLAB** environment using a custom-developed compatibility tool.
- Achieved **100,000 Hz inference** speed by converting the model to **ONNX** and optimizing execution with **ONNX Runtime in C++**, enabling near real-time simulation.

De juillet 2023 à janvier 2024

Data Scientist Internship

Withings Issy-les-Moulineaux

- Enhanced **Deep Learning algorithm** for **heart rate estimation** from PPG signals by adding LSTM layers to the existing architecture, reducing errors **from 14 BPM to 9 BPM** across all activities.
- Analysed and synthesized **latest state of the art**, uncovering novel architectures with features like attention layers, HMM, and spectral data input.
- Enhanced **algorithm robustness** by balancing heart rate references in a new dataset by combining diverse datasets.

De novembre 2022 à juin 2023

System for Detecting Information Manipulation, Student Project

Télécom Paris x Airbus Defence&Space Palaiseau

- Developed a **disinformation-analysis application** on a **tweets** database, providing real-time visualization of content evolution through a Kibana dashboard.
- **Fine-tuned language models** (e.g., RoBERTa) to extract metadata such as sentiment scores and early disinformation indicators, with **containerized** models (Docker) for scalable integration.
- Implemented high-volume data stream from the tweets database using **Kafka**, ensuring reliable ingestion/processing.

Réseaux sociaux

antoinemontagne

antoine-montagne

Langues

Anglais
Operational level

Allemand
Beginner level

Computer Skills

Python
Pandas, Numpy, Polars, SciPy, Scikit-learn, PyTorch, Statsmodels, BeautifulSoup, Optuna, uv

Programming Languages
Python, SQL, C++, Solidity

DevOps
Bash Scripting, Docker, Kubernetes, Spark, GitHub Actions, Terraform

GCP
Vertex AI pipelines, DataProc, BigQuery